Chun-Yuan Chen

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Education

National Sun Yat-Sen University *Kaohsiung, Taiwan*
Master of Science in Electrical Engineering *09/2023 – Present (Expected 05/2026)*

- **Course Highlights:** Design and Analysis of Algorithms (A+), Wireless and Mobile Networks (A+)
- **Exchange Student at Kyushu University** *Fukuoka, Japan*
  - *Graduate School of Information Science and Electrical Engineering* *09/2025 – 03/2026*
    * **Course Highlights:**

National Changhua University of Education *Changhua, Taiwan*
Bachelor of Business Administration in Information Management *02/2020 – 07/2022*

- **Course Highlights:** Big Data Analysis in Practice (A), Databases (A)

National Pingtung University *Pingtung, Taiwan*
Bachelor of Engineering in Computer and Communication *09/2018 – 01/2020*
(Transferred to NCUE to complete degree)

Master Thesis

Handover Strategy for LEO Satellite Networks Based on CTDE Multi-Agent Reinforcement Learning
Propose a handover strategy for LEO satellite networks using multi-agent reinforcement learning to optimize network performance and reduce handover frequency.

- Using Centralized Training with Decentralized Execution (CTDE) framework to train agents representing ground terminals for handover decisions
- Designed reward function considering factors like signal strength, load balancing, and handover costs
- Evaluated performance through simulations, demonstrating improved network throughput and reduced handover frequency compared to traditional methods

Projects

Feature Selection for Yield Prediction in Semiconductor Manufacturing
Performed feature selection and yield prediction on an imbalanced semiconductor manufacturing dataset using logistic regression and cross-validation.

- Designed a reproducible experimental pipeline including preprocessing, feature selection, and cross-validation on real-world industrial data

- Compared filter-based and embedded feature selection methods under a unified evaluation framework using balanced error rate
- Identified potentially informative process variables contributing to yield performance in a complex manufacturing system

Flappy Bird
Utilized Deep Reinforcement Learning to train an AI agent to play Flappy Bird game.

- Using Deep Q Network with Convolutional Neural Network

Transfer packet in TCP
Utilized Deep Q Network algorithm to train an AI agent to play Flappy Bird game.

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Skills

Programming Languages: C++, Python
Tools & Technologies: Pytorch, Scikit-learn, Git, Docker, AWS
Languages: Mandarin (Native), English (Fluent; TOEIC Gold Certificate), Japanese (Basic)